CYS102 Calculus

Assignment 1 (202609)
Part A Group Work

Instructions

(1) Answers must be handwritten in a clear manner.
(2) Use **A4 sized lined** paper.
(3) Write your names, question number and question parts clearly.
(4) Show **complete** working steps to obtain full marks. Marks will be deducted if there are no proper and/or sufficient working steps.
(5) Submit a hardcopy on **3 November 2026** during class.

Question 1 (5 marks)

(a) Given $f(x) = 2x - 3$ and $g(x) = \frac{2}{x}$, evaluate $\frac{f(3)-g(4)}{2}$.

(2 marks)

(b) Solve the following equation for x: $3^{x+1} = 4^{x-1}$.

(3 marks)

Question 2 (7 marks)

(a) What are the one-sided limits? Then, when does the two-sided limit exists?

(2 marks)

(b) With reference to your answer in part (a), determine what type of limit the following is. Then, evaluate the limits.

(i) $\lim_{x\to 3^-}(3x^2 - 2x + 1)$

(2 marks)

(ii) $\lim_{x\to 1}\left(\frac{2-2x^2}{x-1}\right)$

(3 marks)

Question 3 (5 marks)

(a) Find the point(s) of discontinuity for the function $f(x) = \frac{x^2+7x+6}{x^2-4x-5}$.

(1 mark)

(b) Determine whether $f(x) = \begin{cases} x^2 - 9, & x \leq 3 \\ \frac{x^2-x-6}{x-3}, & x > 3 \end{cases}$ is continuous at $x = 3$. Show complete working steps to justify your answer.

(4 marks)

Question 4 (8 marks)

Find $\frac{dy}{dx}$ for the following functions:

(a) $y = \frac{4}{x^2-3}$

(2 marks)

(b) $y = \sqrt[3]{5 - x^4}$

(2 marks)

(c) $y = \sqrt{3x - \sin(4x)}$

(2 marks)

(d) $x^3y^2 - 4x^2 = 1$

(2 marks)

Total marks: 25 marks

- End of Assignment 1 –